

Linguistic-Agnostic Speech Emotion Recognition: A Probing Approach to Acoustic Emotion Encoding

Multimodal Probabilistic Learning of Human Communication Midterm Report

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Abstract

Speech emotion recognition (SER) has advanced significantly in recent years, generally driven by pretrained transformer models such as wav2vec 2.0, HuBERT, and Whisper. These models were originally trained on speech recognition tasks. Because of this, they implicitly extract rich acoustic and linguistic representations. This raises a critical question: during emotion recognition, do these models primarily rely on acoustic prosody, or do they depend on derived linguistic content? In this study, we freeze three pretrained models and probe their internal layers using a logistic regression classifier to identify where emotion-specific information resides within the network architecture. We evaluate this approach across six datasets: RAVDESS (English, acted), EmoDB (German, acted), IEMOCAP (English, spontaneous conversation), SAVEE (English, acted), AESDD (Greek, acted), and MESD (Spanish, acted). Our findings show that middle layers (typically layers 2–5) usually yield optimal performance for acted speech, whereas earlier layers (layers 1–2) are most effective for spontaneous conversation. Furthermore, the experiments show that HuBERT keeps emotion-related information in its deeper layers, whereas wav2vec 2.0 gradually discards it. As these patterns remain consistent across English, German, Greek, and Spanish, we conclude that the emotional encoding within these models is language-agnostic.

Keywords

speech emotion recognition, wav2vec 2.0, HuBERT, whisper, probing, self-supervised learning

1 Problem Definition

Understanding how emotion is encoded in learned speech representations is critical for building generalizable and trustworthy emotion recognition systems. Speech emotion recognition (SER) aims to identify emotional states from speech automatically. The speech signal conveys emotion through two channels: linguistic content (what is said) and paralinguistic cues such as pitch, loudness, and rhythm (how it is said). While both channels contain emotional

information, paralinguistic cues are able to reflect the speaker’s affective state regardless of the words chosen.

Recent pretrained speech transformers such as wav2vec 2.0 [3], HuBERT [4], and Whisper [5] have achieved state-of-the-art SER performance. However, these models are pretrained on large corpora of speech to learn representations that capture spoken language structure and, consequently, encode substantial linguistic information alongside acoustic properties. This raises a fundamental question: do these models predict emotion based on paralinguistic cues, or do they rely on the linguistic content encoded during pre-training? If the latter, their predictions become language-dependent, vulnerable to sarcasm and irony where prosody and lexical meaning diverge, and unreliable when the spoken content does not accurately reflect the speaker’s emotional state.

Our project investigates this question by asking: **at which layers inside these pretrained models is emotion information strongest, and does this pattern hold across different languages, speaking styles, and pretraining paradigms?** Understanding this is a first step toward reducing the influence of linguistic content and building systems that recognize emotion from voice alone. Instead of fine-tuning, we freeze the pretrained models completely and use a simple classifier on each layer to see how much emotion information that layer contains. This probing approach gives us a clear map of where emotion is encoded inside the network, free from any kind of change caused by fine-tuning. To test this thoroughly, our choice of datasets is a critical factor. Each dataset possesses unique characteristics, varying in acting style, script usage, degree of spontaneity, cultural context, and language. Given this diversity, we expect the layer-wise performance to differ across models and datasets.

2 Literature Review

The internal representations of self-supervised speech models have been extensively probed for acoustic and linguistic content, revealing a consistent pattern that motivates our experimental design. Shah et al. [8] trained simple feed-forward networks on the frozen layers of wav2vec 2.0 and found that acoustic features like pitch, energy, and voicing concentrate in the early transformer layers, while linguistic features such as word count, syntax depth, and part-of-speech distributions become increasingly prominent in the middle

and deeper layers. Pasad et al. [10] observed a similar pattern and additionally found that wav2vec 2.0 behaves like an autoencoder in its deeper layers, gradually reverting toward its input and losing phonetic and word-level information. HuBERT, by contrast, does not show this decline. Recent work attributes this difference to HuBERT’s iterative pretraining, which helps it retain more information across all layers [13]. This raises a natural question: does the same pattern apply to emotion, and does HuBERT’s ability to preserve information in deeper layers extend to emotional content as well?

When adapted for speech emotion recognition, transformer models have achieved strong results for predicting arousal, valence, and dominance. Wagner et al. [1] fine-tuned several wav2vec 2.0 and HuBERT variants on MSP-Podcast and obtained state-of-the-art valence prediction without any explicit text input, closing the gap between valence and the other two dimensions. Wang et al. [7] showed that partially fine-tuning only the transformer layers while freezing the CNN yields the best downstream accuracy, suggesting that the transformer layers are where emotion-relevant learning happens. More recently, Goron et al. [9] demonstrated that fine-tuning Whisper achieves superior domain generalization across multiple emotion datasets, outperforming both wav2vec 2.0 and HuBERT in cross-corpus settings. However, Wagner et al. [1] also revealed an important caveat: their best-performing models retained strong valence predictions even on emotionally flat synthetic speech generated from transcriptions alone, suggesting they were relying on linguistic content rather than paralinguistic cues. Triantafyllopoulos et al. [2] confirmed this through controlled probing, showing that valence predictions respond strongly to positive and negative sentiment words and to negations, while arousal and dominance remain unaffected by linguistic manipulations. Since Whisper is explicitly trained for transcription, it is the model most likely to exhibit this text dependence, making it a particularly important case to examine alongside the self-supervised models.

While these studies examined end-to-end performance, a separate line of work has begun probing layer-level representations specifically for emotion content. Singh et al. [14] froze wav2vec 2.0 and HuBERT and classified emotion from each layer across seven datasets in six languages, finding that middle layers consistently outperform both early and late ones. Chiu et al. [15] confirmed this across eleven SSL models, identifying a three-stage pattern: early layers encode acoustics, middle layers peak for emotion, and late layers converge on linguistic content. On the cross-lingual side, Han et al. [16] showed that middle-layer SSL features can match human-level cross-lingual emotion recognition. Building on these findings, our experiments extend the layer-wise probing approach to include Whisper alongside wav2vec 2.0 and HuBERT, use a logistic regression classifier to test whether emotion information is linearly accessible at each layer, and evaluate across four languages and both acted and spontaneous speech to examine how these patterns hold under varied conditions.

3 Data Description

A core part of our experimental design is the selection of datasets. If we evaluated these models on a single English acted corpus, we would not know whether our findings reflect how transformers

process emotion generally, or if we were just capturing the quirks of one specific recording environment. To solve this, we selected six datasets that deliberately span different languages (English, German, Greek, and Spanish), speaking styles (acted versus spontaneous), and recording conditions.

By changing each of these factors, the data will explain the behavior we see in our results. Figure 1 shows the emotion label distributions, and Table 4 in Appendix A provides the exact per-emotion counts.

3.1 RAVDESS (English, Acted)

The Ryerson Audio-Visual Database of Emotional Speech and Song [11] contains 1,440 recordings from 24 professional actors expressing eight emotions. Every actor speaks the exact same two sentences across all emotions, so any successful classification must come from acoustic delivery alone.

3.2 EmoDB (German, Acted)

The Berlin Database of Emotional Speech (EmoDB) consists of 535 German utterances from 10 actors expressing seven emotions in pristine studio conditions. Since wav2vec 2.0 and HuBERT were pretrained exclusively on English, EmoDB lets us test whether emotion encoding transfers to a language the models have never heard.

3.3 IEMOCAP (English, Spontaneous)

The Interactive Emotional Dyadic Motion Capture database [12] contains 5,531 segments of conversational dialogue across four emotion categories. Unlike the acted corpora, IEMOCAP includes unscripted, spontaneous interactions with overlapping speech and subtle emotional shifts, making it the most realistic and challenging dataset in our evaluation.

3.4 SAVEE (British English, Acted)

The Surrey Audio-Visual Expressed Emotion (SAVEE) dataset provides 480 recordings from four male actors speaking British English across seven emotions. SAVEE verifies that the middle-layer peak observed in RAVDESS is not an artifact of North American accents or a specific group of actors.

3.5 AESDD (Greek, Acted)

The Acting Emotional Speech Dynamic Database (AESDD) includes roughly 600 recordings from five actors expressing five emotions in Greek. Because Greek is linguistically distant from both English and German, AESDD serves as a strict cross-lingual test, with both models still peaking in the middle layers at over 90% accuracy.

3.6 MESD (Spanish, Acted)

The Mexican Emotional Speech Database (MESD) contains 648 recordings from 12 Spanish-speaking actors covering six emotions. By adding a fourth language from a distinct Latin American cultural context, MESD further solidifies our cross-lingual findings and provides one of the clearest examples of wav2vec 2.0’s deep-layer accuracy collapse (shown in Figure 4).

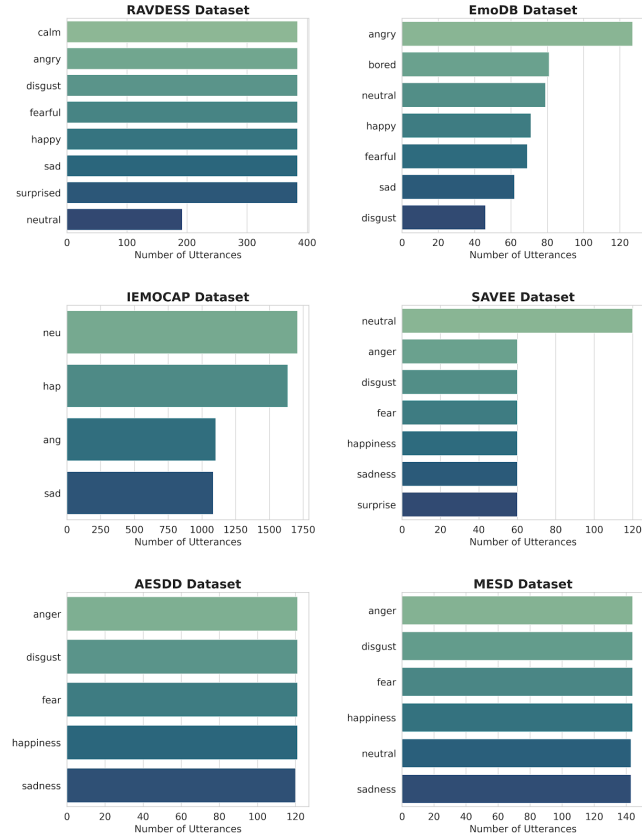


Figure 1: Class distribution of emotional utterances across the six datasets evaluated in this study. IEMOCAP presents the largest volume of data, while EmoDB highlights the greatest natural class variation. Exact counts are provided in Appendix A, Table 4.

4 Method

Our approach follows the probing paradigm used in similar representation analysis [2, 8]. The core idea is straightforward: freeze a pretrained model to keep internal weights unchanged, and then test each layer individually to see how much useful information it holds for a given task. If a layer does well, the emotion information is explicitly available in that representation; otherwise, it is not. This gives us a layer-by-layer map of where emotion is encoded inside the network, free from any kind of change caused by fine-tuning. Figure 2 illustrates this pipeline.

Models. We evaluate three foundation models to cover different architectural and pretraining paradigms:

- **wav2vec 2.0 Base** (facebook/wav2vec2-base-960h, 94M parameters) was pretrained on 960 hours of LibriSpeech using contrastive learning. wav2vec 2.0 learns to predict masked audio segments from the environment context. It has a CNN feature extractor followed by 12 transformer encoder layers, which gives us 13 extractable layers overall.
- **HuBERT Base** (facebook/hubert-base-1s960, 94M parameters) was employed, which is also pretrained on 960 hours of

LibriSpeech but takes a different approach. It first clusters the audio into discrete groups offline, and then trains the model to predict those cluster labels for masked positions. HuBERT shares the same architecture as wav2vec 2.0 (1 CNN + 12 transformer layers), but the difference in training objective creates a substantial effect on how emotion information is distributed across its deeper layers.

- **Whisper Base** (openai/whisper-base, 74M parameters) was trained via large-scale weak supervision on 680,000 hours of multilingual audio paired with transcripts [5]. Its encoder uses 6 transformer layers plus a CNN front-end, giving 7 extractable encoder layers. Because Whisper saw real multilingual speech during its supervised training, it provides a useful reference point for examining how a fully supervised pretraining paradigm shapes internal emotion encoding compared to self-supervised methods. All three models are kept **completely frozen** during all experiments. No weights are updated at any point.

Feature extraction. We resampled each audio file to 16 kHz and passed it through the frozen model with the hidden state output enabled. For wav2vec 2.0 and HuBERT, the model returns hidden states from all 13 layers. For Whisper, we extract from its 7 encoder layers. Each hidden state has a shape of $(1, T, 768)$ or $(1, T, 512)$ depending on the model (where T is the number of time frames). T varies across samples, so we apply mean pooling over the time dimension to produce a single fixed-dimensional vector per audio file per layer. This representation is passed directly to the classifier.

Probing classifier. To measure how much emotion information is in each layer, we train a logistic regression classifier on the pooled features. We chose this linear model specifically because it shows if the information is explicitly available in the representation without needing a complex non-linear model to “find” it. We normalize the features with a standard scaler before training and use 5-fold stratified cross-validation to get more stable results across the different emotion classes. Our final performance is measured by mean accuracy and the weighted F1-score.

Experiments. We evaluated three pretrained architectures (wav2vec 2.0, HuBERT, Whisper) across six structurally diverse datasets (RAVDESS, EmoDB, IEMOCAP, SAVEE, AESDD, MESD). This yields a comprehensive suite of 18 experimental combinations and 198 individual probing evaluations across the varied transformer layers in total. This pipeline allows us to systematically map how acoustic retention differs across pretraining paradigms and languages.

Implementation details. We implemented all experiments in Python using PyTorch and HuggingFace Transformers. Feature extraction ran on an NVIDIA A100-SXM4-40GB GPU. For the probing pipeline, we used Scikit-learn’s StandardScaler for normalization and its LogisticRegression with default parameters.

Pre-processing. wav2vec 2.0 and HuBERT take raw audio waveforms as input directly. Whisper works differently: we first convert each audio file into an 80-channel log-mel spectrogram padded or truncated to 30 seconds, then pass it through the Whisper encoder only. We do not use the decoder at any point. From there, the pipeline is the same across all three models: extract hidden states

from every encoder layer, mean-pool over time, and feed the result into the logistic regression probe.

5 Results

5.1 Comprehensive Layer-Wise Probing

Table 1 presents the peak metrics and optimal layers discovered across all 18 experiments. Figure 5 shows the corresponding full layer-by-layer accuracy curves grouped by model, making the overall topological patterns easy to compare across architectures simultaneously.

On the acted datasets (RAVDESS, EmoDB, SAVEE, AESDD, MESD), wav2vec 2.0 and HuBERT consistently peak in the middle representations (around Layers 2 to 5). RAVDESS reaches exceptionally high accuracy (over 97%) for all models, reflecting its clean studio conditions and highly exaggerated vocal scripts. EmoDB and AESDD similarly reach peak accuracies well over 90%.

However, the behavioral pattern shifts drastically on IEMOCAP: the best layer shifts to the very early representations (Layer 1 or 2) for all three models, and peak accuracy drops to roughly 65% for wav2vec 2.0/HuBERT and 70% for Whisper. Because real conversational emotion is subtler and boundaries between categories are blurrier compared to acted speech, complex internal representation is less effective than early acoustic mapping. Since this dataset only has four categories, we would expect even higher accuracy than the other datasets if the speech were fully acted and the structure was more precise.

Figure 3 illustrates the average layer-wise trajectory across all datasets for each model. The divergence between wav2vec 2.0 and HuBERT is clearly visible: while both models peak together around Layer 4, wav2vec 2.0 drops steeply in the deeper layers, whereas HuBERT stays nearly flat. Table 2 breaks down the exact numerical averages. Figure 4 further visualizes this divergence, showing how consistently wav2vec 2.0 collapses at the deep end while HuBERT and Whisper remain stable. We discuss the structural reasons for this in the key findings below.

5.2 Key Findings

Our analysis of the 18 experimental combinations reveals five primary trends in how transformer layers encode emotional information:

- (1) **Middle layers excel for acted speech.** Across datasets like RAVDESS and EmoDB, performance for wav2vec 2.0 and HuBERT generally peaks between Layers 2 and 5, though HuBERT can peak later (Layer 8 on RAVDESS). Early layers likely only capture raw spectral data (energy and frequency), while middle layers are required to untangle pitch and rhythm. Beyond this point, wav2vec 2.0 accuracy drops because deeper layers prioritize linguistic word recognition over acoustic tone.
- (2) **Spontaneous conversation peaks earlier.** IEMOCAP results differ fundamentally, with the best performance occurring at Layer 1 or 2. Real-world emotion often relies on basic acoustic shifts, like sudden volume changes or pitch drops, which are captured immediately. In contrast, theatrical acting involves stylized contours that require deeper processing to interpret correctly.

Table 1: Best Performing Layer, Peak Accuracy, and F1-Score Grouped by Dataset

Dataset + Model	Best Layer	Acc.	F1
RAVDESS (English, Acted)			
wav2vec 2.0	Layer 5	0.9701	0.9701
HuBERT	Layer 8	0.9757	0.9757
Whisper	Layer 6	0.9750	0.9751
EmoDB (German, Acted)			
wav2vec 2.0	Layer 4	0.9720	0.9719
HuBERT	Layer 5	0.9776	0.9774
Whisper	Layer 5	0.9570	0.9566
IEMOCAP (English, Spontaneous)			
wav2vec 2.0	Layer 1	0.6498	0.6498
HuBERT	Layer 1	0.6597	0.6594
Whisper	Layer 2	0.7009	0.7009
SAVEE (British English, Acted)			
wav2vec 2.0	Layer 5	0.8500	0.8473
HuBERT	Layer 5	0.8750	0.8722
Whisper	Layer 3	0.8208	0.8183
AESDD (Greek, Acted)			
wav2vec 2.0	Layer 2	0.9039	0.9035
HuBERT	Layer 5	0.9403	0.9399
Whisper	Layer 3	0.9072	0.9068
MESD (Spanish, Acted)			
wav2vec 2.0	Layer 2	0.8446	0.8443
HuBERT	Layer 2	0.8597	0.8588
Whisper	Layer 3	0.8109	0.8108

Table 2: Average Probing Accuracy per Layer (wav2vec 2.0, HuBERT, and Whisper across all datasets)

Layer	wav2vec 2.0 Avg.	HuBERT Avg.	Whisper Avg.
CNN	0.8107	0.8017	0.7976
Layer 1	0.8493	0.8427	0.8014
Layer 2	0.8580	0.8582	0.8431
Layer 3	0.8545	0.8621	0.8593
Layer 4	0.8490	0.8608	0.8502
Layer 5	0.8281	0.8609	0.8468
Layer 6	0.8034	0.8471	0.8346
Layer 7	0.7687	0.8240	–
Layer 8	0.7433	0.8000	–
Layer 9	0.7225	0.7789	–
Layer 10	0.6926	0.7946	–
Layer 11	0.6442	0.8199	–
Layer 12	0.5474	0.8064	–

- (3) **HuBERT retains emotion better than wav2vec 2.0.** While wav2vec 2.0 accuracy collapses in deep layers—falling as low as 31% on MESD at Layer 12 (visible in Figure 4)—HuBERT remains stable. This is likely due to their training objectives: wav2vec 2.0’s contrastive task forces deep layers to predict specific spoken words, whereas HuBERT’s cluster-based approach does not overwrite acoustic prosody as aggressively.

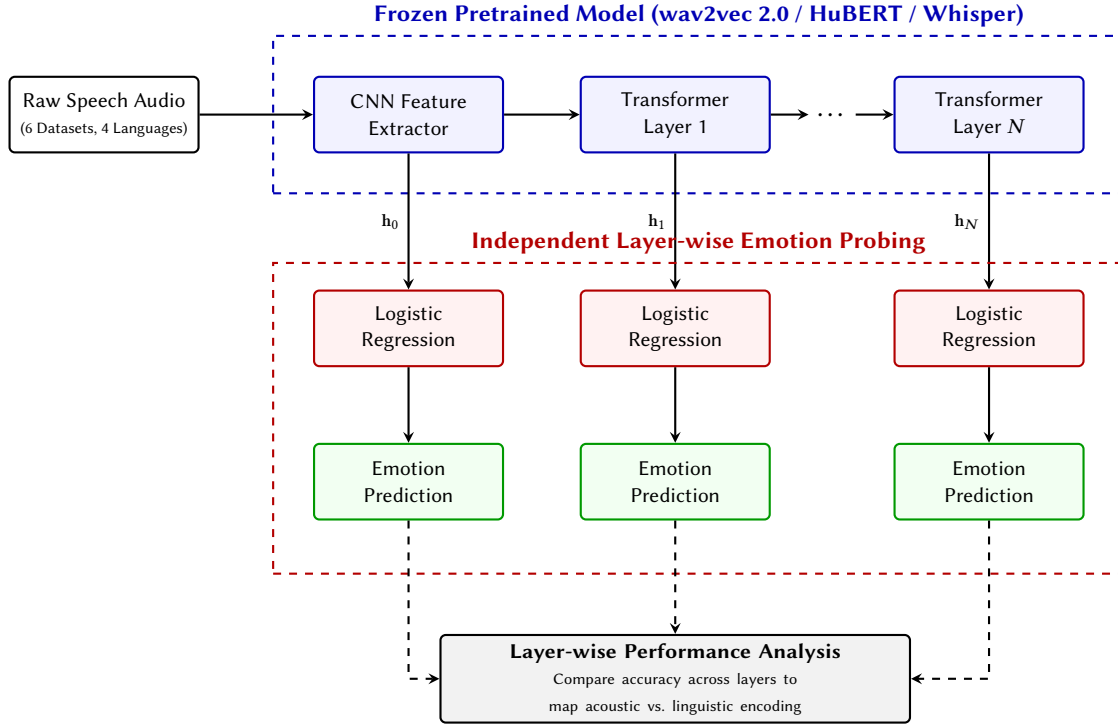


Figure 2: Schematic of the unified layer-wise probing methodology. Each pretrained model is frozen, and hidden state embeddings (h_i) from every layer are independently classified using logistic regression. $N = 12$ for wav2vec 2.0 and HuBERT; $N = 6$ for Whisper.

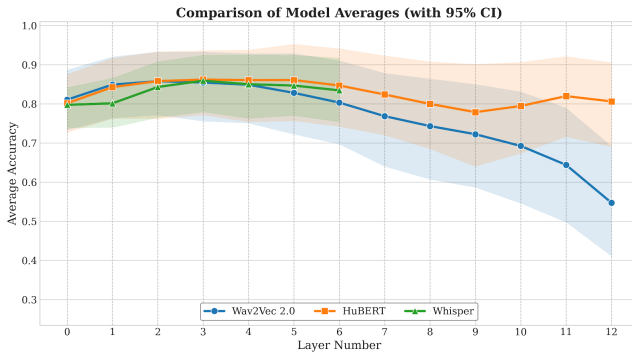


Figure 3: Comparison of average probing accuracy across layers for wav2vec 2.0, HuBERT, and Whisper, computed as the mean accuracy over all six datasets at each corresponding layer.

- (4) **Emotional encoding is language-agnostic.** These representational trends remain identical across English, German, Greek, and Spanish, despite the models never seeing the latter three languages during pretraining. This suggests that the middle layers learn a universal acoustic structure for emotion rather than a language-specific vocabulary mapping.
- (5) **Whisper is highly robust for real-world dialogue.** Despite having fewer encoder layers, Whisper outperformed the other

models on IEMOCAP by roughly 5%. Its massive, supervised training on 680,000 hours of diverse audio likely makes its representations better at handling the noise and overlapping speech found in spontaneous human interactions.

5.3 Per-Emotion Analysis

Figure 6 shows the confusion matrix for Layer 5 of wav2vec 2.0 on RAUDES. Fear is the easiest emotion to recognize; the model classified 37 out of 39 samples correctly. The shaky, high-pitched quality of fear likely makes it stand out clearly to the linear probe. Sadness, however, is the hardest to classify correctly. Only 29 out of 38 samples were correct, and 5 were mistaken for calm. This makes sense because both emotions are quiet and low-energy. They look so similar acoustically that a basic linear classifier simply doesn't have the detail needed to separate them every time. Happy runs into a similar issue: 31 out of 38 are correct, but the mistakes spread across calm, sad, and surprised, which points to the probe struggling most when emotions sit at a similar energy level. On the other hand, angry, disgust, and surprised all come through cleanly because each one has a pretty distinct acoustic signature. What this tells us overall is that the errors are not random. They cluster between emotions that genuinely sound alike, which backs up our broader finding that these representations are picking up on real acoustic structure rather than noise.

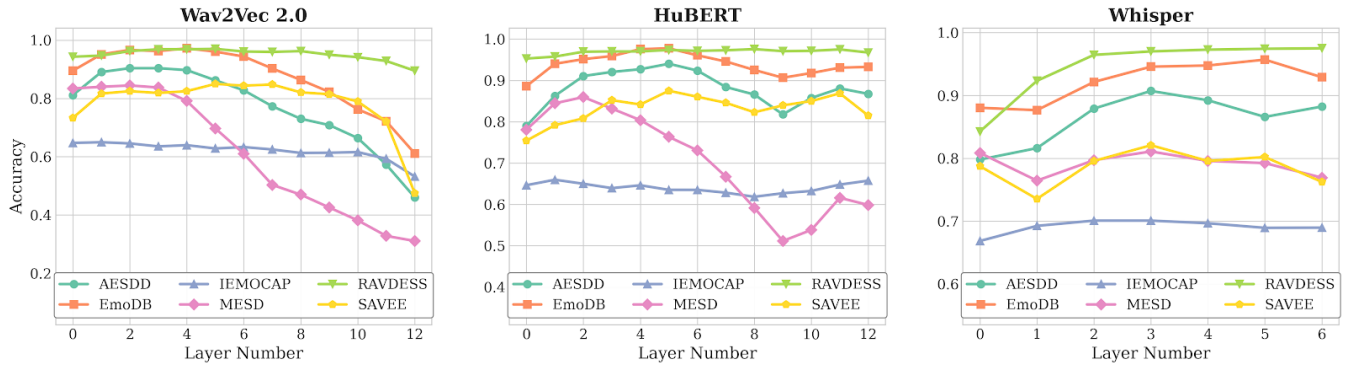


Figure 4: Layer-wise probing accuracy broken down by model, showing per-dataset curves and a mean accuracy line for each. Left: wav2vec 2.0 exhibits a characteristic deep-layer decline. Center: HuBERT retains emotion information into the final layers. Right: Whisper maintains stable accuracy across its shorter 7-layer encoder block.

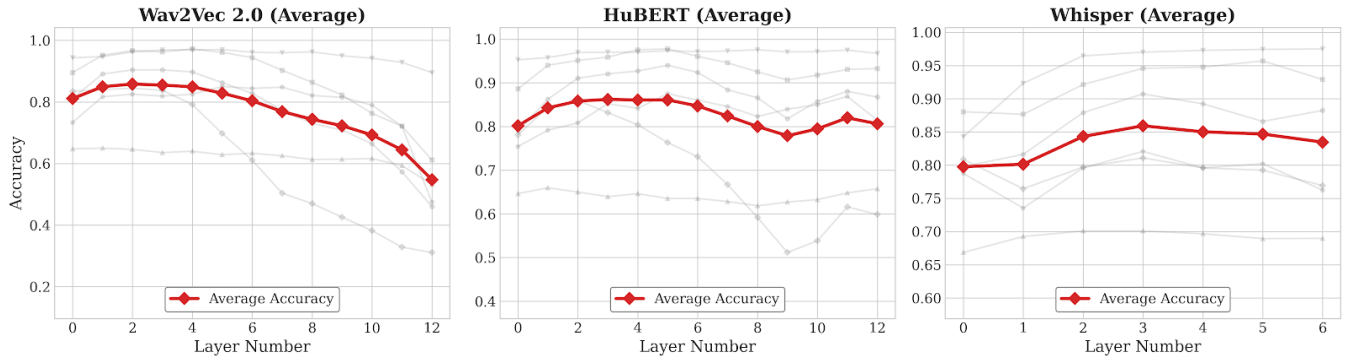


Figure 5: Layer-wise probing accuracy for all 18 experiments simultaneously, grouped by model. Each color represents a distinct dataset (consistent across panels). wav2vec 2.0 and HuBERT each evaluate 13 extractable layers; Whisper evaluates 7.

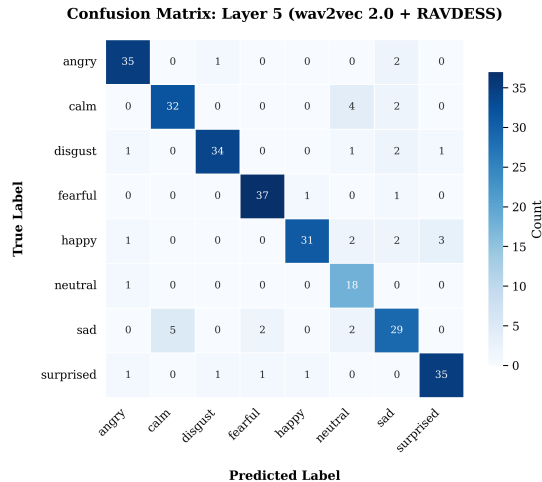


Figure 6: Confusion matrix from Layer 5, wav2vec 2.0 on RAVDESS. Rows are actual emotions, columns are predictions.

5.4 Cross-Model Correlation

To see if these architectures encode emotion in the same way, we calculated the Pearson correlation (r) between their layer-by-layer accuracy profiles. For the wav2vec 2.0 and HuBERT comparison, we used all 13 layers. Since Whisper only has 7 encoder layers, we matched by absolute layer index against the first 7 layers of the other models, noting that this compares Whisper’s full depth against only the early-to-middle range of the deeper architectures. The results across all datasets are detailed in Table 3.

The numbers reveal three clear patterns. First, HuBERT and Whisper process acted speech very similarly, showing high correlations on RAVDESS ($r = 0.942$), AESDD ($r = 0.880$), and EmoDB ($r = 0.830$). Second, wav2vec 2.0 and HuBERT only strongly align on the Spanish MESD dataset ($r = 0.932$). On the other acted datasets, their correlation is much lower. This directly reflects our earlier finding: wav2vec 2.0’s accuracy collapses in its deep layers, while HuBERT’s accuracy stays flat, causing their profiles to diverge.

Finally, the consistency completely breaks down on the spontaneous IEMOCAP dataset. Across the board, correlations drop to near-zero or become negative (for example, $r = -0.113$ between wav2vec 2.0 and HuBERT). This tells us that while the models share similar internal pathways for processing stylized, acted emotion,

Table 3: Pearson Correlation (r) of Layer-Wise Accuracy Profiles

Dataset	wav2vec 2.0 vs. HuBERT	wav2vec 2.0 vs. Whisper	HuBERT vs. Whisper
RAVDESS	0.221	0.887	0.942
EmoDB	0.488	0.673	0.830
IEMOCAP	-0.113	-0.216	0.032
SAVEE	0.275	0.001	0.342
AESDD	0.384	0.522	0.880
MESD	0.932	0.461	0.187

they handle the messy, subtle cues of real-world conversation in entirely different ways. For the final part of the project, we will try to use other datasets and augmentation to evaluate this idea even further.

6 Plans to Finish the Project

For the final submission, we have several directions for extending and strengthening the analysis:

- (1) **Evaluate using additional models.** We plan to add WavLM Base+ and MERT as additional models, representing other self-supervised training paradigms. We are also exploring parameter-efficient fine-tuning with LoRA to study how adapting just a small number of weights changes the layer-wise emotion profile compared to our completely frozen probes.
- (2) **Noise augmentation.** Following related work, we plan to use the ESC-50 environmental sound dataset to add realistic background noise at varying signal-to-noise ratios. This will reveal if the models still rely on the exact same layers when the acoustic signal becomes messy.
- (3) **BERT alignment analysis.** We will compare each layer’s hidden states against BERT text embeddings to directly measure how much a layer resembles a linguistic representation versus an acoustic one. This will give us mathematical proof for our theory that wav2vec 2.0 shifts toward text processing in its deeper layers.
- (4) **Speaker-independent evaluation.** We will conduct a leave-one-speaker-out cross-validation to confirm that our accuracy results are not inflated by speaker identity leakage. This guarantees the model is truly recognizing universal emotions, rather than just memorizing how specific actors sound.
- (5) **Learned layer mixer.** Instead of selecting a single best layer, we will extract representations from multiple layers simultaneously and learn an attention-based mixing network on top. The goal is to let the system dynamically weight layers that carry strong acoustic cues while downweighting layers that have shifted toward linguistic processing, directly addressing our core finding that emotion and language information occupy different depth regions.
- (6) **Adversarial training.** We will experiment with adversarial training by adding a loss penalty whenever the acoustic representation looks too similar to text. If successful, this technique

could directly fix the text-bias flaw we observed in architectures like wav2vec 2.0.

- (7) **Congruent vs incongruent evaluation.** We will test whether model predictions break down when the spoken text and the emotional delivery do not match. For instance, we want to see if the model gets confused when a speaker says a positive word but uses a clearly angry voice.
- (8) **MSP Podcast evaluation.** We will evaluate on the MSP Podcast corpus, the largest publicly available speech emotion dataset, to test whether our early-layer advantage for spontaneous speech holds on a much larger real-world corpus beyond IEMOCAP.
- (9) **Large-scale compiled benchmark.** We plan to explore aggregated benchmarks like Speechcraft that unify multiple datasets, letting us stress-test whether our cross-lingual patterns hold when dataset-specific quirks are averaged out across a larger pool of speakers and conditions.

7 Team Contributions

(1) Nima Kelidari:

- **Initiation & Development:** Co-authored the proposal, conducted the problem definition and literature review, and initiated the project on GitHub.
- **Validation & Debugging:** Debugged the data pipeline and validated experiments across USC’s CARC and Google Colab.
- **Data Visualization:** Redesigned all plots, diagram schemas, and tables throughout the document to ensure clear, intuitive cross-model comparisons.
- **Writing & Formatting:** Rewrote earlier report sections (Abstract, Problem Definition, Literature Review, Data Description), managed report formatting, and ensured document consistency and referencing across the entire report.

(2) Minoo Ahmadi:

- **Initiation & Development:** Co-authored the project proposals (multiple iterations due to topic changes).
- **Experiments:** Extended the wav2vec 2.0 and HuBERT experiments to three additional datasets, then extended all six datasets with Whisper experiments. Re-ran and validated all model-dataset combinations to ensure correctness, and generated the layer-wise accuracy plots and model comparison figures.
- **Writing & Revision:** Revised and rewrote the Problem Definition and Literature Review sections of the report. Revised the midterm presentation slides.
- **Project Guidance:** Contributed to shaping the overall research direction and giving guidance throughout the project.

(3) Chaitanya Parwatkari:

- **Pipeline Development:** Built the complete layer-wise probing framework from scratch, which extended across other dataset combinations, covering hidden state extraction with mean pooling, StandardScaler normalization, and 5-fold stratified cross-validation.
- **Experiments & Correlation Analysis:** Ran initial core experiments across wav2vec 2.0 and HuBERT on 3 datasets, producing per-layer accuracy curves, performance tables, the

confusion matrix, and the comparison plots used in the report. Implemented the cross-model correlation analysis comparing how different architectures encode emotion across layers.

- **Writing & Project Direction:** Wrote the initial Problem Definition, Method, Results, Key Findings, Per-Emotion Analysis, and Cross-Model Correlation sections. Defined the project roadmap in the Plans section, outlining concrete next steps for the final submission.
- **Report Finalization & Presentation:** Verified all prose numbers against updated experiment tables, fixed broken cross-references and caption mismatches across multiple revision rounds. Designed and structured the 10-slide midterm presentation.

(4) **Xiangxu Lin:**

- **Report Initiation:** Initiated the midterm report, provided the template to follow the outline, and set up the document in ACM Latex format.
- **Writing:** Wrote the initial Data Description section for all six datasets in the report.

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A Detailed Dataset Class Distributions

The count of each dataset class is separated and detailed in Table 4.

Table 4: Detailed class distributions of emotional utterances across all six datasets used in this study.

Dataset	Emotion	Count
RAVDESS	Neutral	192
	Calm, Angry, Disgust, Fearful, Happy, Sad, Surprised	384 (ea.)
EmoDB	Angry	127
	Bored	81
	Neutral	79
	Happy	71
	Fearful	69
	Sad	62
	Disgust	46
IEMOCAP	Neutral (neu)	1708
	Happy (hap)	1636
	Angry (ang)	1103
	Sad (sad)	1084
SAVEE	Neutral	120
	Anger, Disgust, Fear, Happiness, Sadness, Surprise	60 (ea.)
AESDD	Anger, Disgust, Fear, Happiness	121 (ea.)
	Sadness	120
MESD	Anger, Disgust, Fear, Happiness	144 (ea.)
	Neutral, Sadness	143 (ea.)

Note: Where “(ea.)” appears, the stated count applies to each emotion listed in the adjacent cell. Emotions with identical sample sizes have been grouped to keep the table readable in two-column format. The IEMOCAP counts reflect the standard four-class evaluation subset (anger, happiness/excitement, neutral, sadness) used in most SER benchmarks, not the full unmerged label set. Datasets recorded with controlled actor scripts, such as RAVDESS and SAVEE, naturally show near-perfect class balance, whereas EmoDB and IEMOCAP exhibit more variation due to their less constrained recording procedures.